

Arsenic Presence in the Cuyama Valley Groundwater Basin

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Changes to groundwater flow alter groundwater quality. Arsenic, a natural and hazardous contaminant, can leach from aquifer sediments and mobilize into water supplies due to excessive pumping (1). The U.S. Environmental Protection Agency established a maximum contaminant level (MCL) of 10 $\mu\text{g/L}$ for arsenic to limit carcinogenic health risks from long-term exposure (2). In Cuyama Valley, arsenic is localized in the central region, where pumping is greatest (Fig. 1; 3). Since 2015, Well CA 4210009_002_002 accounts for 106 samples, while the remaining 12 wells total 17 samples, reflecting nonrepresentative monitoring (4). Although concentrations frequently exceed the MCL, the Cuyama Groundwater Sustainability Plan lacks arsenic sustainability thresholds and recommends sampling only once every five years, limiting its ability to protect local water supplies (3).

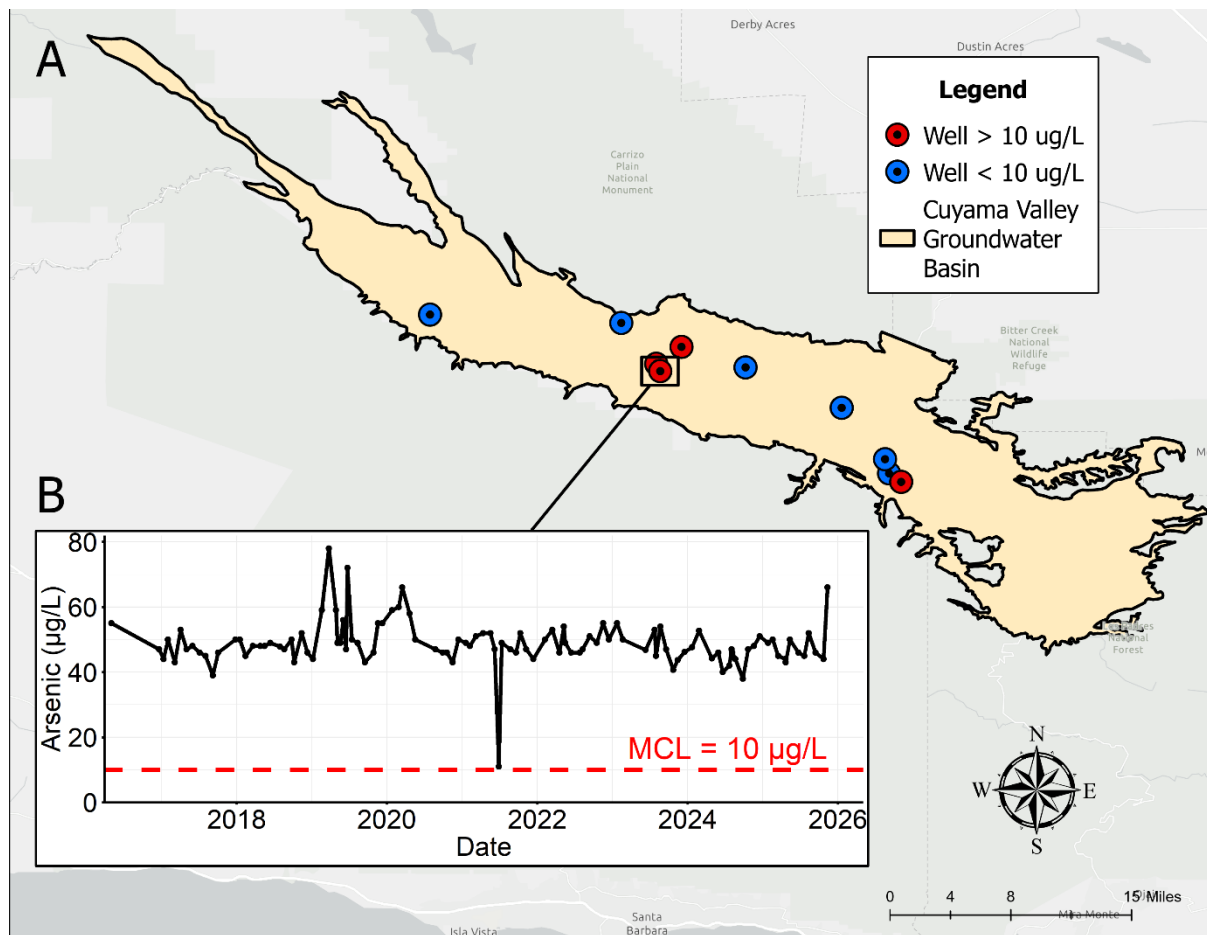


Fig. 1. Spatial distribution of arsenic occurrences in the Cuyama Valley Groundwater Basin (2015-2025). (A) Locations of monitoring wells classified by compliance with the U.S. Environmental Protection Agency maximum contaminant level (MCL) of 10 $\mu\text{g/L}$. Wells which have exceeded the MCL are shown in red. Wells below the MCL are shown in blue. (B) Time series of arsenic concentrations for Well CA 4210009_002_002 from 2015 to 2025. The dashed line represents the MCL. Arsenic concentrations in Central Cuyama frequently and substantially exceed the MCL. Data acquired from the Groundwater Ambient Monitoring and Assessment Program (4).

References

1. R. Smith, R. Knight, S. Fendorf, Overpumping leads to California groundwater arsenic threat. *Nat. Commun.* **9**, 2089 (2018).
2. U.S. Environmental Protection Agency, National primary drinking water regulations: Chemical contaminant rules. <https://www.epa.gov/dwreginfo/chemical-contaminant-rules>. Accessed 18 February 2026.
3. Cuyama Basin Groundwater Sustainability Agency, "Cuyama Valley Groundwater Basin Final 2025 Groundwater Sustainability Plan" (Cuyama Basin Groundwater Sustainability Agency, Santa Barbara County, CA, 2025).
4. California State Water Resources Control Board, GAMA - Groundwater Ambient Monitoring and Assessment Program. <https://gamagroundwater.waterboards.ca.gov>. Accessed 10 February 2026.